FAB FLOW | Semiconductor OHT Routing Optimization

Synthetic FAB Experiment · Conference Portfolio / Learners × YPICS Seminar Preparation

Executive Summary / 연구 요약

반도체 FAB에서 여러 OHT가 같은 레일과 교차로를 공유하면 최단경로만으로는 전체 운송 효율을 설명하기 어렵다. 48개 노드와 82개 단방향 링크를 가진 가상 FAB 모델을 구성하고, 전체 경로 예약과 다음 구간만 예약하는 제어기를 비교하였다.

Main study: 600 runs (3 scenarios × 5 policies × 40 matched seeds). Safety sensitivity: 360 additional runs. Synthetic requests and synthetic topology; no factory MES or real sensor linkage.

Problem & Model

A full-route lock preclaims every node and edge along a route. A partial reservation grants only the next section when destination occupancy and prospective wait-cycle constraints permit. Wait-for graph edges describe a blocked vehicle waiting for another; a cycle flag is a warning rather than physical deadlock certification.

Reinforcement Learning / 강화학습

Q-learning records action values for reduced representations of location, destination, demand condition and activity. Q(s,a) is updated with a learning rate of 0.22 and a discount factor of 0.87. The trained policy is compared with an otherwise similar untrained policy to isolate learning effects. A safe-action shield limits path detours and disallows reserved intersections.

Evaluation & Statistics

All policies in a scenario receive the same generated job requests for 40 paired evaluation seeds. Paired mean differences, 95% bootstrap confidence intervals and Holm-adjusted exact sign tests support the comparisons. Completed-only transport lead times may be biased because unfinished jobs are excluded.

Key means / 평균 완료 운송 (25분)

Scenario	Full-route	Partial dynamic	Trained Q
normal	18.925	24.450	24.300
surge	20.900	34.325	34.900

closure	18.600	24.475	24.500

Interpretation and limitations

Partial dynamic routing improved completion counts over conservative full-route reservation across all three tested conditions: +5.525 (normal), +13.425 (surge), and +5.875 (rail closure) mean jobs. Learning contributed +0.575 mean jobs only in surge (95% CI [+0.200,+1.000], Holm-adjusted p=0.0369); other conditions did not establish a significant learning effect.

모형에 정의한 노드 중복 점유 위반 0건과 보호장치의 순환대기 신호 감소는 실제 장비에 대한 충돌 방지 인증이 아니다. FAB 동역학, 레일 블록, 장비 지연, 실시간 센서 동기화, 생산 스케줄링, 다중 네트워크 일반화는 아직 추가 검증이 필요하다.

Full source, readme, paired inference CSV and interactive schematic:
github.com/doyunsin883-debug/semiconductor-oht-routing-optimization / fabflow-oht-phase3-deadlock.vercel.app